Vector Autoregression

Chapter 7

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Outline

- Structural and reduced form VARs
- Lag selection
- Granger causality
- SVAR identification
- Impulse response functions
- Forecast error and variance decomposition
- VARX and I(1) variables

Structural and reduced form VARs

Section 1

Structural and reduced form VARs

- SARIMAX models have **one dependent** variable
- They accept exogenous regressors X_t; where X is taken as given, with **no feedback** from y to X
- Growth (g) and inflation (π) are jointly determined: each affects the other
- We need a system in which **every** variable is endogenous

Structural and reduced form VARs

- Assume simple Phillips curve model
- This is the **structural** (theoretical) model
  - $g_t = \alpha_{10} - \alpha_{12}\pi_t + \beta_{11}g_{t-1} + \beta_{12}\pi_{t-1} + \varepsilon_t^g$
    $\pi_t = \alpha_{20} - \alpha_{21}g_t + \beta_{21}g_{t-1} + \beta_{22}\pi_{t-1} + \varepsilon_t^\pi$
  - Within one period
    - $\Delta\varepsilon_t^g \rightarrow \Delta g_t \rightarrow \Delta\pi_t$ (through α_{21})
    - $\Delta\varepsilon_t^\pi \rightarrow \Delta\pi_t \rightarrow \Delta g_t$ (through α_{12})
  - Dynamic effects
    - $\Delta\varepsilon_t^g \rightarrow \Delta g_t \rightarrow \Delta\pi_t \rightarrow \Delta g_{t+1} \rightarrow \Delta\pi_{t+1} \ldots$
    - $\Delta\varepsilon_t^\pi \rightarrow \Delta\pi_t \rightarrow \Delta g_t \rightarrow \Delta\pi_{t+1} \rightarrow \Delta g_{t+1} \ldots$

Structural and reduced form VARs

- These models cannot be estimated through OLS
  - $\rho(g_t, \varepsilon_t^{\pi}) \neq 0 \wedge \rho\left(\pi_t, \varepsilon_t^{g}\right) \neq 0$
- Model is **not identified** without restrictions
  - Structural parameters = 10: 8 coefficients + 2 shock variances
  - Reduced-form moments = 9: 6 mean coefficients + 3 residual variances and covariance
- Solution: Build the **reduced** form and apply restrictions
  - One equation per **endogenous** variable, each a function of **lagged** endogenous variables only
  - The regressors are predetermined, so each equation can be estimated by OLS
- IMPORTANT! The meaning of the coefficients change (see next slide)

Structural and reduced form VARs

- Get the **reduced** form
  - $g_t = \alpha_{10} - \alpha_{12}\pi_t + \beta_{11}g_{t-1} + \beta_{12}\pi_{t-1} + \varepsilon_t^g$
    $\pi_t = \alpha_{20} - \alpha_{21}g_t + \beta_{21}g_{t-1} + \beta_{22}\pi_{t-1} + \varepsilon_t^\pi$
  - $\begin{bmatrix} 1 & \alpha_{12} \\ \alpha_{21} & 1 \end{bmatrix} \begin{bmatrix} g_t \\ \pi_t \end{bmatrix} = \begin{bmatrix} \alpha_{10} \\ \alpha_{20} \end{bmatrix} + \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} \begin{bmatrix} g_{t-1} \\ \pi_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^g \\ \varepsilon_t^\pi \end{bmatrix}$
  - $A_0 \begin{bmatrix} g_t \\ \pi_t \end{bmatrix} = \begin{bmatrix} \alpha_{10} \\ \alpha_{20} \end{bmatrix} + A_1 \begin{bmatrix} g_{t-1} \\ \pi_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^g \\ \varepsilon_t^\pi \end{bmatrix}$
  - $\begin{bmatrix} g_t \\ \pi_t \end{bmatrix} = A_0^{-1} \begin{bmatrix} \alpha_{10} \\ \alpha_{20} \end{bmatrix} + A_0^{-1} A_1 \begin{bmatrix} g_{t-1} \\ \pi_{t-1} \end{bmatrix} + A_0^{-1} \begin{bmatrix} \varepsilon_t^g \\ \varepsilon_t^\pi \end{bmatrix}$
  - $\boldsymbol{y_t = \Gamma_0 + \Gamma_1 y_{t-1} + \tilde{\varepsilon}_t}$

Structural and reduced form VARs

- The VAR(p) model
  - Each equation has p lags of **each** endogenous variable $(n = 3)$

$$y_{1t} = \gamma_{10} + \sum_{l=1}^{p} \gamma_{11l} y_{1t-l} + \sum_{l=1}^{p} \gamma_{12l} y_{2t-l} + \sum_{l=1}^{p} \gamma_{13l} y_{3t-l} + \tilde{\varepsilon}_{1t}$$

  - $y_{2t} = \gamma_{20} + \sum_{l=1}^{p} \gamma_{21l} y_{1t-l} + \sum_{l=1}^{p} \gamma_{22l} y_{2t-l} + \sum_{l=1}^{p} \gamma_{23l} y_{3t-l} + \tilde{\varepsilon}_{2t}$

$$y_{3t} = \gamma_{30} + \sum_{l=1}^{p} \gamma_{31l} y_{1t-l} + \sum_{l=1}^{p} \gamma_{32l} y_{2t-l} + \sum_{l=1}^{p} \gamma_{33l} y_{3t-l} + \tilde{\varepsilon}_{3t}$$

  - Every equation has the same regressors, so equation-by-equation OLS is efficient
  - The scalar notation gets heavy as n and p grow, so we use matrix algebra ☺
    - $\boldsymbol{y_t = \Gamma_0 + \Gamma_1 y_{t-1} + \Gamma_2 y_{t-2} + \cdots + \Gamma_p y_{t-p} + \tilde{\varepsilon}_t}$

Structural and reduced form VARs

- Parameters and roots
  - Each equation has np lagged regressors plus an intercept: $np + 1$ ($n = 3$, $p = 2$, gives 7)
  - Mean parameters: $n(np + 1) = n^2p + n = 21$; add $n(n+1)/2 = 6$ covariance parameters
  - Growth is **quadratic in n and linear in p**. With ~260 quarterly observations, a VAR(6) uses 57 mean parameters.
  - Stability
    - A VAR(p) model has **np eigenvalues** (6 here)
    - All must lie inside the unit circle
  - Lag selection is key
    - Too few lags leave autocorrelated residuals. Too many waste degrees of freedom.

Lag selection

Section 2

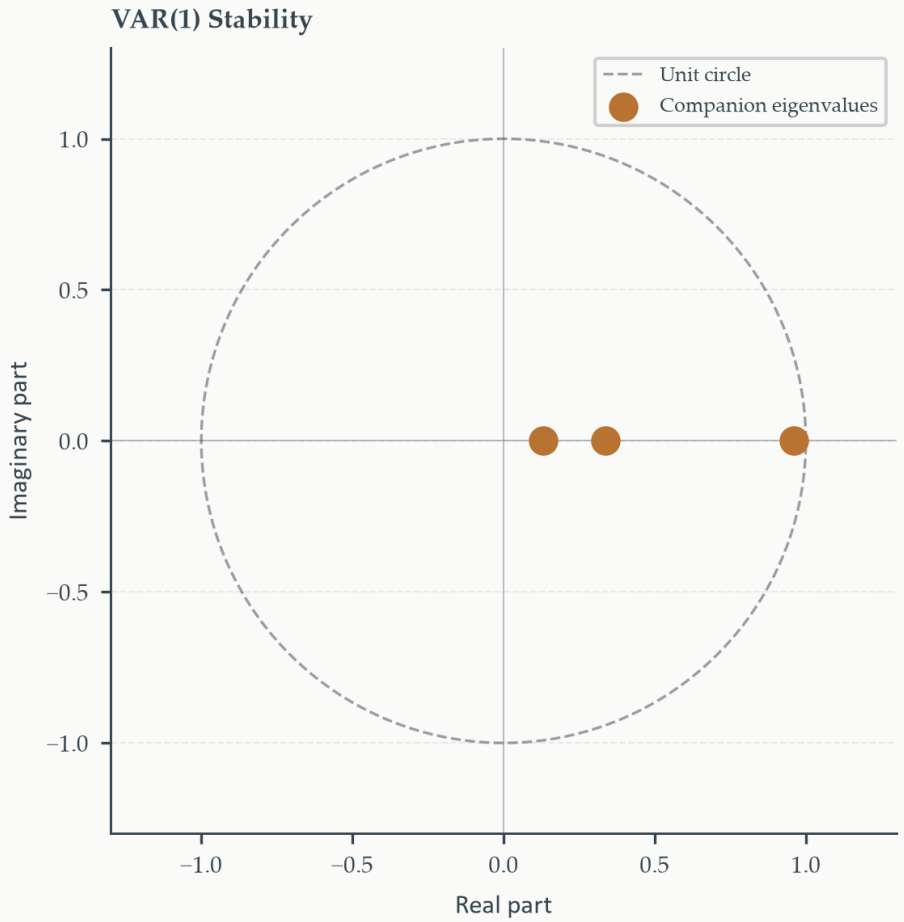
Lag selection

- Choose $p_{\max}$ (here, $p_{\max} = 6$)
- Estimate VAR(1), …, VAR(6) on a **common sample**, so the IC are comparable
  - AIC, BIC, HQC
- If IC agree, first guess is a “strong” start
- If IC disagree
  - Start with a candidate p
  - Test for serial correlation
  - If they fail, raise p
- Residual analysis **overrides** IC

Lag selection

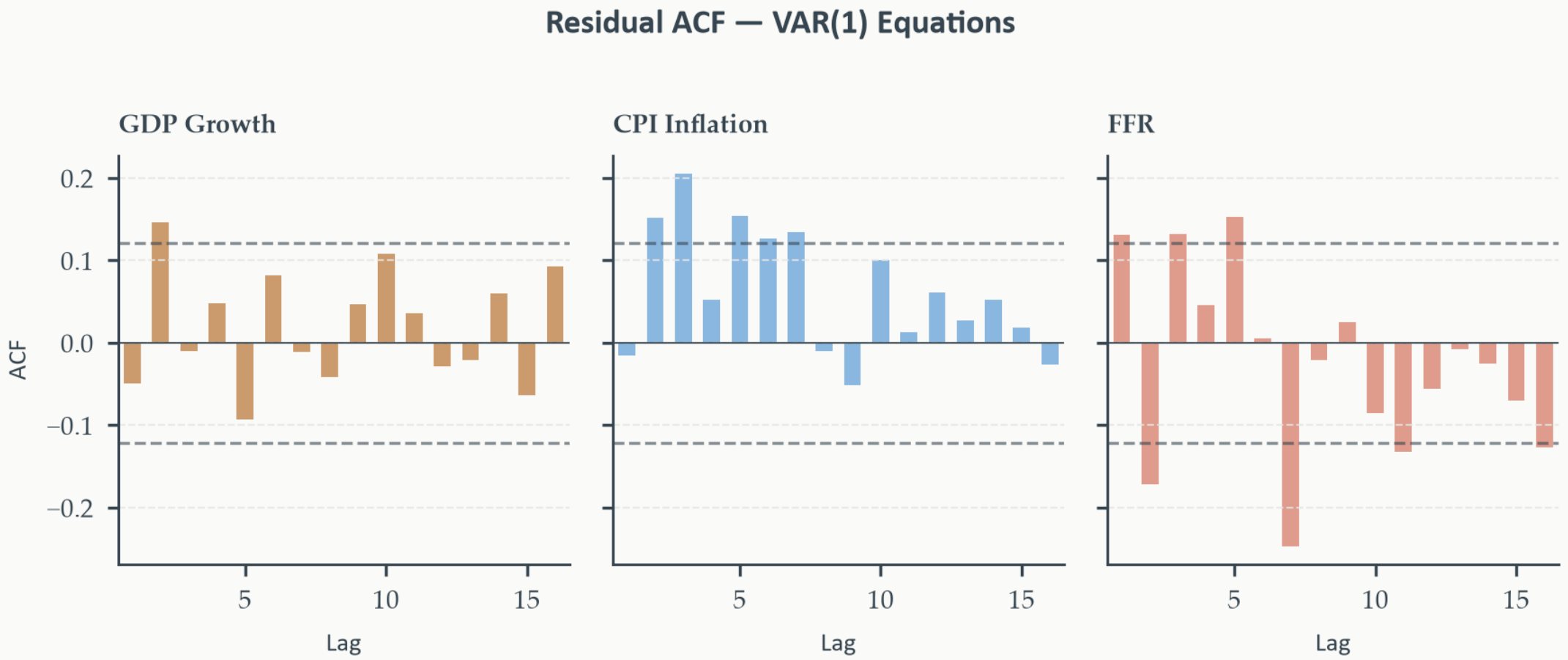
p	AIC	BIC	HQC
1	1.821	1.985*	1.887
2	1.731	2.019	1.847
3	1.635	2.047	1.801*
4	1.680	2.217	1.896
5	1.612*	2.275	1.879
6	1.617	2.406	1.934

Lag selection

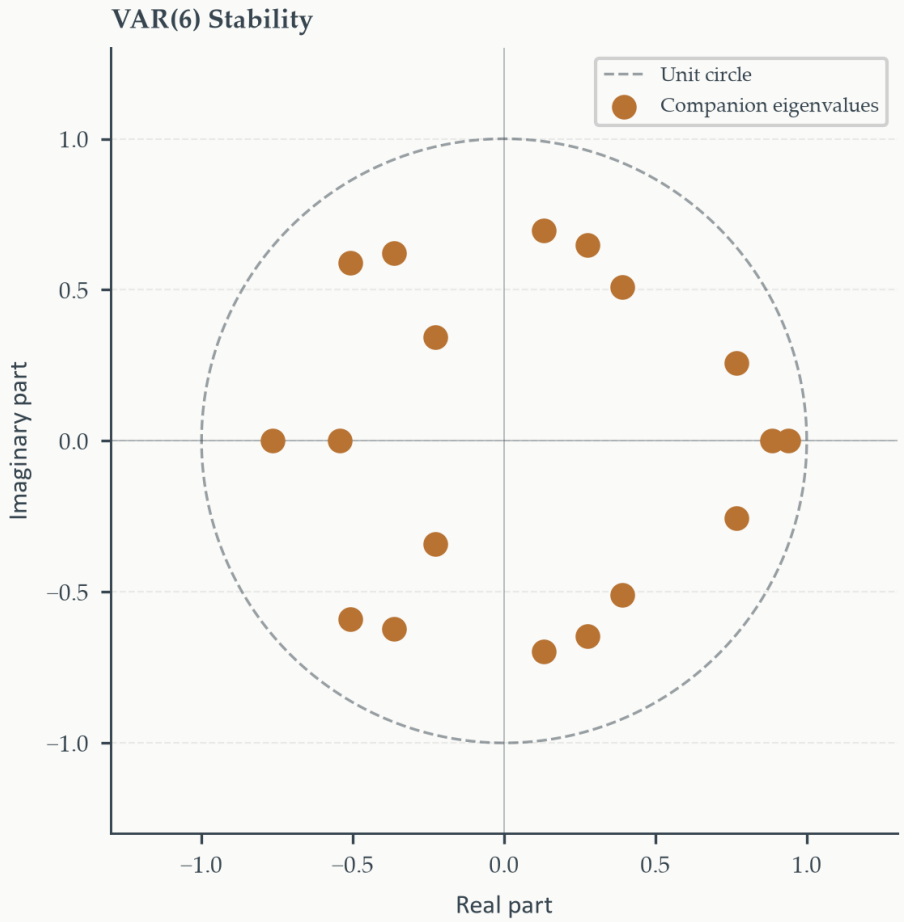


- The VAR(1) has $np = 3$ companion eigenvalues, all real, at about 0.13, 0.34, and 0.96
- All inside the unit circle, so the VAR model is stable
- The largest eigenvalue (≈ 0.96) is close to the boundary; very persistent dynamics
- Stability is necessary, but not sufficient; look at the residuals next.

Lag selection

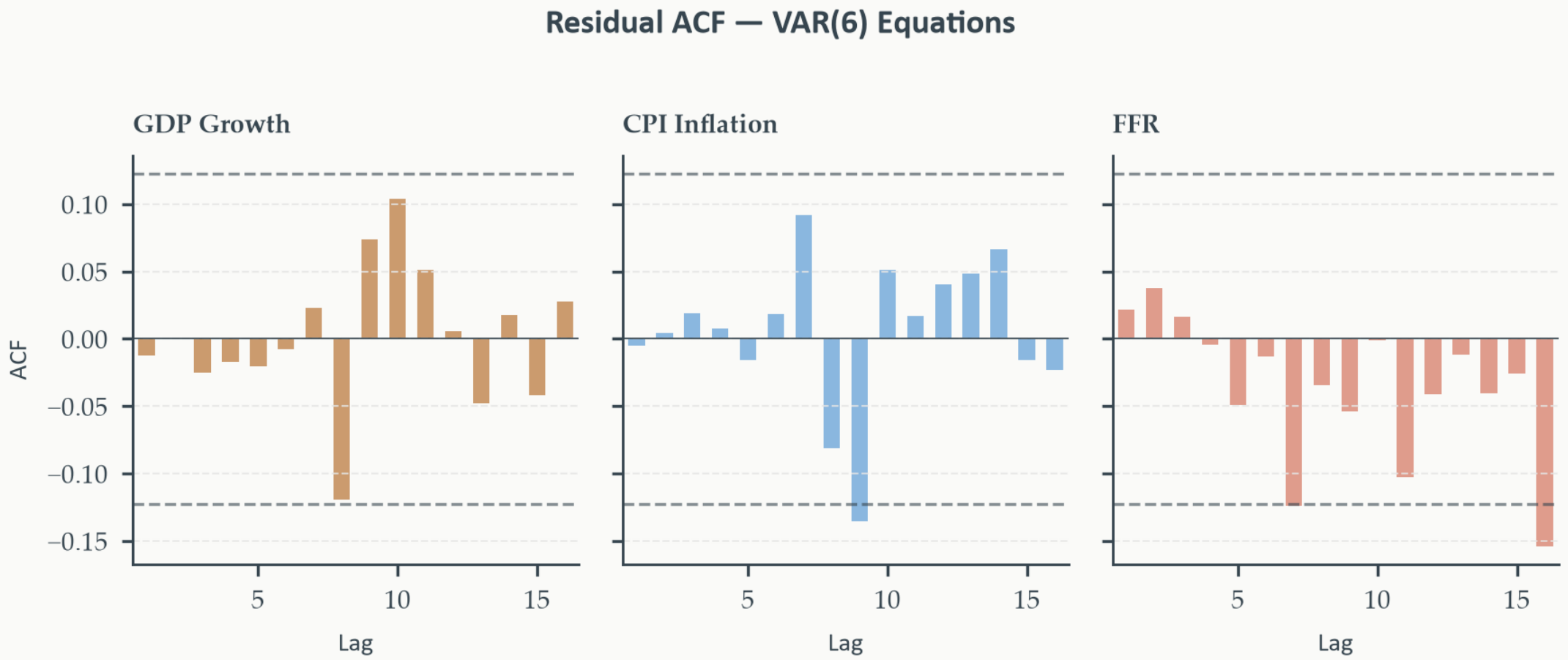


Lag selection



- The VAR(6) has $np = 18$ companion eigenvalues, 4 real and 7 complex-conjugate pairs
- All inside the unit circle, so the VAR model is stable with its largest modulus (≈ 0.94)
- Roots/eigenvalues dampen oscillations (cycles)

Lag selection



Granger causality

Section 3

Granger causality

- Granger causality
  - Statistical **precedence**, not structural (causal) causality
    - A falling barometer Granger-causes storms: it helps predict them, but it does not cause them
    - Likewise, asset prices help predict GDP because markets *anticipate* output, not because they cause it
  - y_j **Granger-causes** y_i if past values of y_j help predict y_i **beyond what its own past and the past of all other variables in the system** can predict
  - Equivalently: adding y_j's lags lowers the forecast MSE of y_i

Granger causality

Cause		Effect	p-value	
GDP Growth	→	CPI Inflation	0.3959	
GDP Growth	→	FFR	0.0000	***
CPI Inflation	→	GDP Growth	0.4592	
CPI Inflation	→	FFR	0.0488	**
FFR	→	GDP Growth	0.0000	***
FFR	→	CPI Inflation	0.0008	***

H_0: row cause does not Granger-cause column effect.

*** p<0.01 ** p<0.05 * p<0.10

Granger causality

- Reading the Granger-causality results
  - Monetary policy **reaction function**
    - $g \to i \quad (p < 0.001)$
    - $\pi \to i \quad (p = 0.049)$
    - Consistent with a Taylor rule: The Fed responds to output and inflation
  - Monetary policy effects
    - $i \to g \quad (p < 0.001)$
    - $i \to \pi \quad (p < 0.008)$
    - The Fed is forward-looking. It moves in anticipation of output and inflation, so predictive content is not necessarily a policy effect
  - Not rejected
    - Non-rejection does not mean zero effect: The test has limited power and is blind to indirect channels $(g \to i \to \pi)$

SVAR identification

Section 4

SVAR identification

- Each reduced-form residual is a **mixture** of all structural shocks
  - $\tilde{\varepsilon}_t = A_0^{-1}\varepsilon_t$
- Therefore, reduced-form residuals are correlated ($\Sigma_{\tilde{\varepsilon}}$ is not diagonal)
  - A "shock to $\tilde{\varepsilon}^{\pi}$ alone" is not a well-defined experiment
- We need to recover structural shocks
- Counting: $\Sigma_{\tilde{\varepsilon}}$ has $n(n+1)/2$ distinct elements
  - The structural side has n^2 unknowns
  - We need $n(n-1)/2$ additional restrictions
- Where do restrictions come from?
  - Zero restrictions (Cholesky), sign restrictions, long-run restrictions, external instruments

SVAR identification

- Cholesky decomposition
  - Assumptions: contemporaneous effects are **recursive**. Order the variables from most to least contemporaneously exogenous: $g \rightarrow \pi \rightarrow i$
    - $A_0 = \begin{bmatrix} 1 & 0 & 0 \\ \alpha_{21} & 1 & 0 \\ \alpha_{31} & \alpha_{32} & 1 \end{bmatrix}$ → a lower-triangular matrix
      - g_t does not react to π_t or i_t within the quarter
      - π_t reacts to g_t but not to i_t, within the quarter
      - i_t reacts to both g_t and π_t, within the quarter
    - The three zeros above the diagonal are $n(n-1)/2$ required restrictions, so the system is **exactly identified**
    - Implication for IRFs: The first-ordered shock can move every other variable. The last-ordered shock moves only its own variable
    - Where does the ordering come from? Economic theory

SVAR identification

- Sign restriction
  - Idea: replace zero restrictions with **signs on impulse responses** to a structural shock, over some horizon (h)
  - Let P be a factor of Σ: $PP' = \Sigma$
  - Let Q be a factor if I: $QQ' = I$
  - Then, there is a $\tilde{P} = PQ$ such that $\tilde{P}\tilde{P}' = PQQ'P' = \Sigma$
  - Data cannot tell the difference between P and $\tilde{P}$
  - Cholesky is the point $Q = I$.

SVAR identification

- Sign restriction (cont…)
  - Procedure
    - Estimate the VAR(p) model and compute P (Cholesky)
    - Draw a random matrix $Z \sim N(0, I_n)$. Take $Z = QR$ (QR decomposition) and keep Q, with the signs of R's diagonal fixed positive
    - Compute $\widetilde{\Theta}_h = \Theta_h Q$ for $h = 0, \ldots, H$
    - Keep the Q if the signs hold, otherwise discard it
    - Repeat until you have many accepted draws
    - Report them median and percentile bands across accepted draws

Impulse response functions

Section 5

Impulse response functions

- An impulse response answers: if structural shock j hits at t, how does variable i move at $t + h, h = 0, 1, 2, \ldots$?
  - $\varepsilon_t^g \rightarrow \{g_{t+h}, \pi_{t+h}, i_{t+h}\}$
  - With $n = 3$, there are $n^2 = 9$ response functions
  - Shocks must be uncorrelated to be a clean experiment

Impulse response functions

- Step **1**: Estimate the covariance matrix
  - $\Sigma_{\tilde{\varepsilon}} = \mathbb{E}[\tilde{\varepsilon}\tilde{\varepsilon}'] \approx \frac{\hat{\tilde{\varepsilon}}\hat{\tilde{\varepsilon}}'}{T-k}$
  - $\mathbb{E}[\varepsilon\varepsilon'] = I$ is the unit-variance normalization of **structural** shocks
- Step **2**: Apply the Cholesky decomposition to $\Sigma_{\tilde{\varepsilon}}$
  - $\Sigma_{\tilde{\varepsilon}}$ is positive definite, so a unique lower-triangular matrix P exists
- Step **3**: Solve for P such that $PP' = \Sigma_{\tilde{\varepsilon}}$

Impulse response functions

- Step **3** (numerical example)
  - $\Sigma_{\tilde{\varepsilon}} = \begin{bmatrix} 4.0 & 1.2 \\ 1.2 & 9.0 \end{bmatrix}; P = \begin{bmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{bmatrix}; P' = \begin{bmatrix} p_{11} & p_{21} \\ 0 & p_{22} \end{bmatrix};\ PP' = \Sigma_{\tilde{\varepsilon}}$
  - $\begin{bmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} p_{11} & p_{21} \\ 0 & p_{22} \end{bmatrix} = \begin{bmatrix} 4.0 & 1.2 \\ 1.2 & 9.0 \end{bmatrix}$
  - $\begin{bmatrix} p_{11}^2 & p_{11}p_{21} \\ p_{11}p_{21} & p_{21}^2 + p_{22}^2 \end{bmatrix} = \begin{bmatrix} 4.0 & 1.2 \\ 1.2 & 9.0 \end{bmatrix}$
    - $p_{11}^2 = 4$
    - $p_{11}p_{21} = 1.2$
    - $p_{21}^2 + p_{22}^2 = 9$

Impulse response functions

- Step **3** (numerical example) (cont…)
  - $P = \begin{bmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 2.0 & 0 \\ 0.6 & 2.9 \end{bmatrix}$
    - $p_{11} = 2.0 \rightarrow$ the standard deviation of y_1 (GDP growth) structural shock
    - $p_{21} = 0.6 \rightarrow$ the impact on π of a one standard deviation output shock
    - $p_{22} \approx 2.9 \rightarrow$ the standard deviation of the inflation shock, net of what the output shock already explains

Impulse response functions

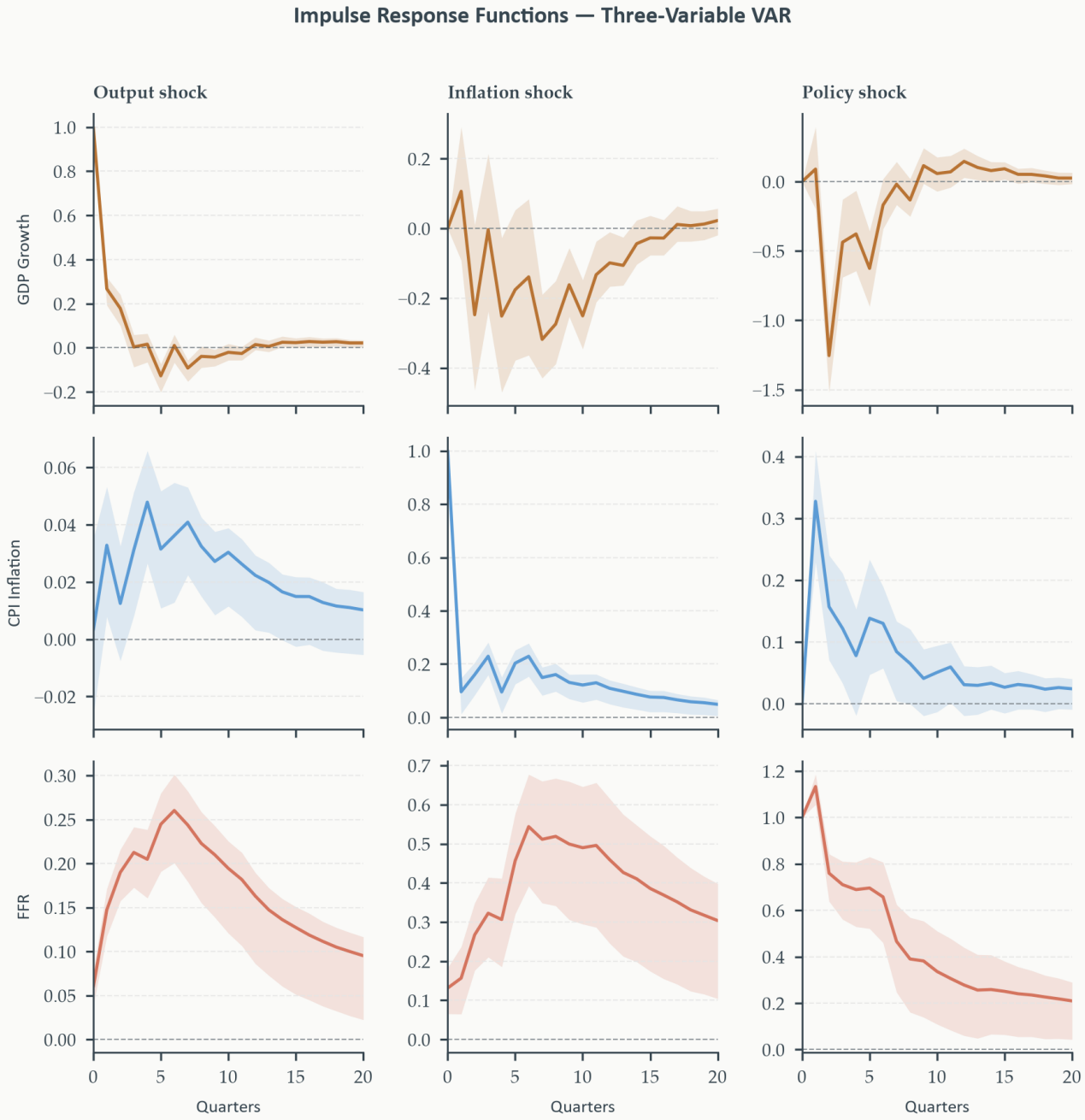
- Step $\boldsymbol{4}$ – assume a VAR(1) model
  - Let $\boldsymbol{e} = (1,0)'$ be a one-standard-deviation output shock
  - On $t = 0$ (contemporaneous shock)
    - $\Theta_0 e = Pe$
    - $\Theta_0 e = [2.0 \quad 0.6]'$
    - Column j is the impact factor of shock j. The first-ordered shock can move every variable on impact. The last shock moves only its own variable
  - On $t = 1$ (next period)
    - $\Theta_1 e = \Gamma_1 Pe$
  - On horizon h (VAR(1) only)
    - $\Theta_h e = \Gamma_1^{\mathrm{h}} Pe$

Impulse response functions

- Step **4** (cont…)
  - Full IRF
    - $y_t = \Gamma_0 + \sum_{h=0}^{\infty} \Theta_h \varepsilon_{t-h}$
    - $\Theta_h = JF^h J' P$
    - If the VAR model is stable: $F^h \to 0$ and IRF effects decay $y_t \to \bar{y}$
      - F is a "bookkeeping" matrix: It stacks the current and lagged values (Γ_i) into one system.
      - J picks out the current values from the stacked matrix

Impulse response functions

- Step 5
  - Estimate the VAR model and keep $\hat{\Gamma}_0, \hat{\Gamma}_1, \ldots, \hat{\Gamma}_p, \hat{\tilde{\varepsilon}}_t$
  - Resample the residuals $(\tilde{\varepsilon}_t^*)$ using $\hat{\tilde{\varepsilon}}_t$
  - Rebuild the data recursively: $y_t^* = \hat{\Gamma}_0 + \hat{\Gamma}_1 y_{t-1} + \cdots + \hat{\Gamma}_p y_{t-p} + \tilde{\varepsilon}_t^*$
  - Restimate and recompute P^* and Θ_h^*
  - Repeat 500+ times (assumes roughly i.i.d. residuals) and extract the $\pm$ one-standard-deviation band: the 16th and 84th percentiles (not a 95% interval)
  - Compute the band at each horizon h



- Columns are shocks
- Row are responses (Cholesky ordering)
- Each column is scaled so the own response on impact equals 1 (not one standard deviation)

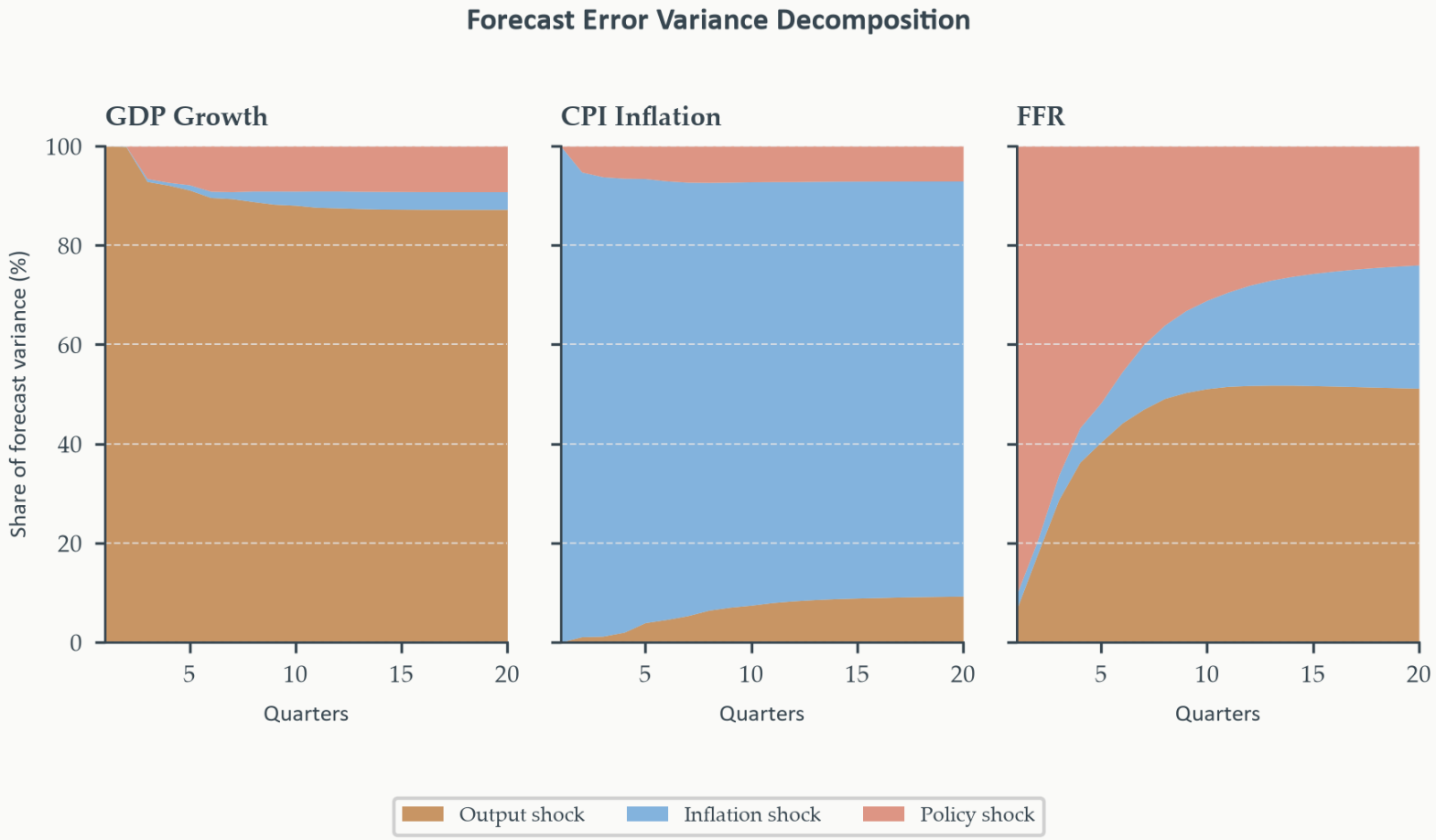
Forecast error and variance decomposition

Section 6

Forecast and variance decomposition

- IRF
  - *How* a variable responds to a shock, at each horizon
- FEVD (forecast error variance decomposition)
  - *How much* of the forecast uncertainty comes from each shock
  - Like the IRFs, FEVD depends on the Cholesky ordering
  - At h = 1 the first-ordered variable (GDP growth) is entirely its own shock

Forecast and variance decomposition



VARX and I(1) variables

Section 7

VARX and I(1) variables

- VARX = VAR + exogenous variables X_t
  - $y_t = \Gamma_0 + \sum_{l=1}^{p} \Gamma_l y_{t-l} + \sum_{j=0}^{s} B_j X_{t-j} + \tilde{\varepsilon}_t$
  - X is **exogenous**: it influences y but y does not influence X
  - X has no equation, so there is no IRF or FEVD
    - Report **dynamic multipliers**
  - Why include lags of X?
    - Effects of X on y are typically delayed
    - Choose p and s by IC on a common sample

VARX and I(1) variables

- I(1) variables
  - First, test each variable for stationarity
  - Do not difference a series that is already stationary
  - Exogenous X
    - If $X \sim \mathrm{I}(0)$: include in levels
    - If $X \sim I(1)$ and not cointegrated with y: Include in first difference
    - If $X \sim I(1)$ and cointegrated with y: model long run relationship explicitly (VECM model, next chapter)
  - Endogenous y
    - All $I(0)$: VAR in levels
    - $I(1)$ and not cointegrated: VAR in first differences
    - $I(1)$ and cointegrated (VECM, next chapter)

Vector Autoregression

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